

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) Let m be the Lebesgue measure on $[0, 1]$ and $I_A(x)$ denote the indicator function of A .

(a) Consider for $n \geq 1$, $f_n(x) = nI_{[0, \frac{1}{n}]}(x)$ and $f(x) = 0$ for $x \in [0, 1]$. Show:

(i) $f_n \rightarrow f$ a.e.(m).

(ii) f_n is **not** uniformly integrable (UI).

(iii) Check $f_n \rightharpoonup f$ in L^1 .

(b) In general, under what conditions on the functions $\{f_n\}$ and f , would

" $f_n \rightarrow f$ a.e. (m)" imply " $f_n \rightarrow f$ in L^p ($0 < p < \infty$)"?

(c) Show that $f_n \rightarrow f$ in L^p implies that $f_n \rightarrow f$ in L^q , for any $0 < q < p < \infty$.

((3×6)+6+8=32 points)

- (2) Suppose X is a random variable defined on some probability space $(\Omega, \mathcal{F}, P)$.

(a) If $M(t) = E(e^{tX})$, $t \in \mathbb{R}$ is the moment generating function of X , then show

$$P(X > a) \leq \inf_{t>0} e^{-ta} M(t), \quad a > 0.$$

(b) If $X \geq 0$ is random variable with $Var(X) < \infty$. Prove:

$$P(X > 0) \geq \frac{[E(X)]^2}{E(X^2)}.$$

[Hint: Use Cauchy-Schwarz Inequality]

(8+10=18 points)

- (3) (a) **State** precisely the *Dominated convergence Theorem (DCT)* (Not the EDCT!).

(b) **Prove** that the conclusion of DCT remains true even if " $f_n \rightarrow f$ a.e. (μ)" is replaced by " $f_n \rightarrow f$ in measure" in the statement.

(6 + 9 = 15 points)

- (4) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, μ_1, μ_2, ν are σ -finite measures on it. Define the set function: $\mu_3 \equiv \mu_1 + \mu_2$ (i.e., for all $A \in \mathcal{F}$, $\mu_3(A) = \mu_1(A) + \mu_2(A)$). Then, prove the following:

(a) μ_3 is a measure.

(b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$.

(c) Express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).

(d) If $\mu \ll \nu$ and $\mu \perp \nu$ holds, show that μ is the zero measure.

(e) If $\mu_1 = \text{Lognormal}(5, 6)$, $\mu_2 = \text{Geometric}(0.3)$ and $\nu = \text{Poisson}(8)$ on $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then find the Lebesgue decomposition of μ_3 w.r.t. ν , and find the Radon Nikodym derivative $\frac{d\mu_{3a}}{d\nu}$, where μ_{3a} is the absolutely continuous component of μ_3 in the decomposition.

(6+8+8+8+10=40 points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^1 p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1, n \in \mathbb{N}$	$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b]) / (b - a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx, a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha, \beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0, 1)}(x) m(dx),$ where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b), a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2 / 2\sigma^2}}{x} I_{(0, \infty)}(x) m(dx)$